

Letter to the Editor

Mediterranean diet, mild cognitive impairment, and Alzheimer's disease

Dear Sir,

We read with great interest the recent article by [Steele and colleagues \(2006\)](#), which reviewed recent evidence on the suggested role of specific foods and diets thought to lower the risk of developing Alzheimer's disease (AD). This concept is further supported by recent evidence that certain diets have been associated with a lower incidence of AD ([Panza et al., 2004](#)). In fact, antioxidants, dietary fatty acids, and micronutrients appear to have a role, and evidence is at least suggestive that diets rich in fruits and vegetables and other dietary approaches may permit a beneficial effect on the risk of dementia ([Solfrizzi et al., 2003](#)). In the last years, the study approach was to associate single micro- or macronutrients to age-related cognitive decline, mild cognitive impairment (MCI), AD, or vascular dementia ([Del Parigi et al., 2006](#)). In this picture, several hallmarks of the Mediterranean diet were linked to increased risk or with a protective effect against cognitive impairment ([Solfrizzi et al., 1999, 2006a,b](#)). The typical dietary pattern of Mediterranean diet is characterized by high intakes of vegetables, fruits and nuts, legumes, cereals, fish, and monounsaturated fatty acids (MUFA); relatively low intakes of meat, and dairy products, and moderate consumption of alcohol. In fact, higher levels of consumption of olive oil are considered the hallmark of the traditional Mediterranean diet. In particular, MUFA, consequently to the high consumption of extra-virgin olive oil, represent the most important fat in Mediterranean diet. Cumulative evidence suggests that extra-virgin olive oil may have a role in the protection against cognitive decline ([Solfrizzi et al., 1999, 2006a](#)), other than against coronary disease and several types of cancer because of its high levels MUFA and polyphenolic compounds ([Trichopoulou et al., 2003](#)). [Steele and colleagues \(2006\)](#) cited in their article findings from the Italian Longitudinal Study on Aging (ILSA) demonstrated that dietary MUFA intake was significantly associated with a reduction in the odds ratio of developing cognitive impairment ([Solfrizzi et al., 1999; Panza et al., 2006](#)). The findings of this study were recently confirmed in a 8.5-year follow-up, in which high MUFA, polyunsat-

urated fatty acids (PUFA), and total energy intake were significantly associated with a better cognitive performance in time ([Solfrizzi et al., 2006a](#)). Finally, recent findings from the ILSA demonstrated that while dietary fatty acids intakes were not associated with incident MCI, high PUFA intake appeared to have borderline non-significant trend for a protective effect against the development of MCI ([Solfrizzi et al., 2006b](#)).

Very recently, [Scarmeas and colleagues \(2006\)](#) reported the results of a community-based study involving 2258 nondemented individuals in New York, in which adherence to a traditional Mediterranean diet was associated with significantly reduction in risk for AD. However, in the study of [Scarmeas and colleagues \(2006\)](#), the median daily intake of MUFA to SFA ratio for individual food categories by Mediterranean diet score tertiles was <1 and in overall about 2.5 times lower than the same value calculated from other studies on Mediterranean diet ([Solfrizzi et al., 2006c](#)). [Scarmeas and colleagues \(2006\)](#) used in this report a scale indicating the degree of adherence to the traditional Mediterranean diet originally constructed by [Trichopoulou and colleagues \(1995\)](#) and revised to include fish intake ([Hu et al., 2002](#)). A value of 0 or 1 was assigned to each of nine indicated components with the use of the sex-specific median as the cut-off. The total Mediterranean-diet score (MeDi) ranged from 0 (minimal adherence to the traditional Mediterranean diet) to 9 (maximal adherence). The wide diffusion of this methodological approach exploring possible associations between dietary habit, in particular Mediterranean diet, and several outcomes in more and more epidemiological areas (cancer, cardiovascular disease, or dementia) has undeniable advantages but some concern it should be admit. In fact, several devices related to the use of dietary composite score should be considered, in evaluating the effects of nutrient intakes in different groups (low and high consumption) in unbiased manner ([Solfrizzi et al., 2006d](#)). In particular, a variation in group heterogeneity, which could be caused by selection of cut-offs, can change in predictable ways the reliability and validity of the scale indicating the degree of adherence to a particular dietary pattern. The effects of change in observed-score variance on reliability can be estimated, if it is assumed that error variance remains constant after selection ([Solfrizzi et al., 2006d](#)). On the basis of these considerations, it is

not difficult to guess that a “whole-diet approach” such as MeDi could not be readily transferred to other populations, included study population traditionally eating according to Mediterranean dietary pattern.

The findings of Scarmeas and colleagues (2006) and those from the ILSA (Solfrizzi et al., 1999, 2006a,b) strongly suggested that further observational studies on cognitive decline need to be implemented on the adherence to a specific dietary pattern rather than association studies with single nutrients. If successfully replicated, randomized clinical trials should ideally follow to specifically address the question of whether a dietary pattern may have a role in the prevention of cognitive impairment. A recent study by Small and colleagues (2006), in people with mild age-related memory complaints, demonstrated the positive effects of a 14-day healthy longevity lifestyle program on word fluency, and activity in the left dorsolateral prefrontal cortex at [fluorine-18] fluorodeoxyglucose (FDG) positron emission tomography (PET) scans in comparison of the control. Cardiovascular conditioning and brief relaxation exercises designed to lower stress are recommended each day. Suggested shopping lists and menus guide subjects to follow a healthy diet plan, including five daily meals emphasizing antioxidant fruits and vegetables, n-3 PUFA, and low glycemic index carbohydrates (Small et al., 2006). The conceptual basis for the healthy diet plan suggested in the study of Small and colleagues was that diets high in n-3 PUFA from olive oil or fish (Solfrizzi et al., 1999, 2006a,b), as well as those rich in antioxidant fruits and vegetables (Solfrizzi et al., 2003; Del Parigi et al., 2006), are associated with less age-related cognitive decline. It is also possible that moderate lifestyles in general, which obviously vary according to different cultural environments, protect from cognitive deterioration. Thus, it may not be the direct effect of diet or specific nutrients that provide the protection, but healthy diets very similar to the Mediterranean dietary pattern may be an indicator of a complex set of favourable social and lifestyle factors. Future longitudinal studies will determine the long-term effects of such interventions, whether they eventually lower the risk for developing cognitive impairment, and the weight of a specific dietary pattern in these combined healthy lifestyle programs.

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Francesco Panza ^{a,*}
Cristiano Capurso ^b
Alessia D’Introno ^a
Anna M. Colacicco ^a
Angelo Del Parigi ^c
Gaetano Gagliardi ^a
Giovanni Breglia ^a
Antonio Capurso ^a
Vincenzo Solfrizzi ^a

^a Department of Geriatrics, Center for Aging Brain, Memory Unit, University of Bari, Bari, Italy

^b Department of Geriatrics, University of Foggia, Foggia, Italy

^c The John B. Pierce Laboratory, New Haven, CT, USA
E-mail address: geriat.dot@geriatria.uniba.it (F. Panza).

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* Corresponding author. Tel.: +39 080 5473685; fax: +39 080 5478860.